HEALTHCARE PATIENT CHURN ANALYSIS REPORT

Prepared from the provided Healthcare EDA notebook, patient dataset, cleaned dataset, and Power BI dashboard.
Date: 28 August 2026

1. Executive Summary

The analysis covers 2,000 patients and focuses on patient churn, engagement, attendance behavior, satisfaction, billing friction, geography, digital portal usage, insurance, and referral behavior. The overall patient churn rate is 68.35% (1,367 patients), while 633 patients are retained.

- Overall churn is high at 68.35%, making retention the central business problem.
- Billing issues are associated with a higher average churn rate: approximately 77% with an issue versus 68% without one.
- Distance is a strong observed churn risk: churn rises to about 72.65% for patients living more than 30 miles away.
- The largest engagement segment is Moderately engaged (947 patients), followed by Engaged (520).
- The At Risk segment contains 453 patients, about 22.7% of the patient base.
- Overall satisfaction is not strongly related to out-of-pocket cost in the EDA, so cost alone should not be treated as the primary satisfaction driver.
- Wait-time satisfaction is relatively consistent across specialties, supporting a system-wide rather than specialty-specific operational improvement approach.
- Female and male churn rates are almost identical (68% and 69% in the notebook calculation), indicating no meaningful gender gap in churn.

2. Project Objective

The objective was to explore the healthcare patient dataset, identify patterns associated with patient churn and engagement, engineer a composite engagement score, create meaningful patient segments, and present the resulting business insights through a Power BI dashboard.

- Understand the structure and quality of the patient data.
- Explore demographic, geographic, operational, financial, satisfaction, and engagement patterns.
- Engineer standardized behavioral scores and an overall engagement score.
- Identify high-risk patient groups and operational friction points.
- Build a dashboard that allows the analysis to be explored using interactive filters.

3. Data Source and Dataset Overview

The original patient_churn_dataset.csv contains 2,000 records and 21 fields. The analysis produced updated healthcare.csv with 29 fields after feature engineering.

Item	Result
Original dataset	2,000 rows × 21 columns
Updated dataset	2,000 rows × 29 columns
Unique patient IDs	2,000
Missing values	0 across all original columns
Duplicate rows	0
Churned patients	1,367
Retained patients	633
Overall churn rate	68.35%

3.1 Main Data Fields

Field	Purpose
PatientID	Unique patient identifier
Age / Gender / State	Patient demographic and geographic attributes
Tenure_Months	Length of relationship with the clinic
Specialty	Medical specialty associated with the patient
Insurance_Type	Insurance/payment category
Visits_Last_Year	Annual visit frequency
Missed_Appointments	Number of missed appointments
Days_Since_Last_Visit	Recency of patient interaction
Overall_Satisfaction	Overall patient satisfaction score
Wait_Time_Satisfaction	Satisfaction with waiting time
Staff_Satisfaction	Satisfaction with staff

Provider_Rating	Provider rating
Avg_Out_Of_Pocket_Cost	Average patient out-of-pocket cost
Billing_Issues	Indicator for billing issues
Portal_Usage	Digital patient portal usage indicator
Referrals_Made	Number of referrals made
Distance_To_Facility_Miles	Patient distance from facility
Churned	Target indicator: 1 = churned, 0 = retained

4. Data Preparation and Cleaning

1. Loaded the patient dataset using pandas.
2. Inspected the dataset using head(), columns, describe(), and info().
3. Checked missing values using isnull().sum(); all original columns contained zero missing values.
4. Checked for duplicate records; the dataset contained zero duplicate rows.
5. Standardized column names to lowercase for easier analysis and Power BI usage.
6. Created additional analytical features: visit_score, recency_score, attendence_score, tenure_score, engagement_score, engagement_category, distance_bins, and age_group.
7. Exported the resulting 29-column dataset as updated healthcare.csv.

5. Feature Engineering

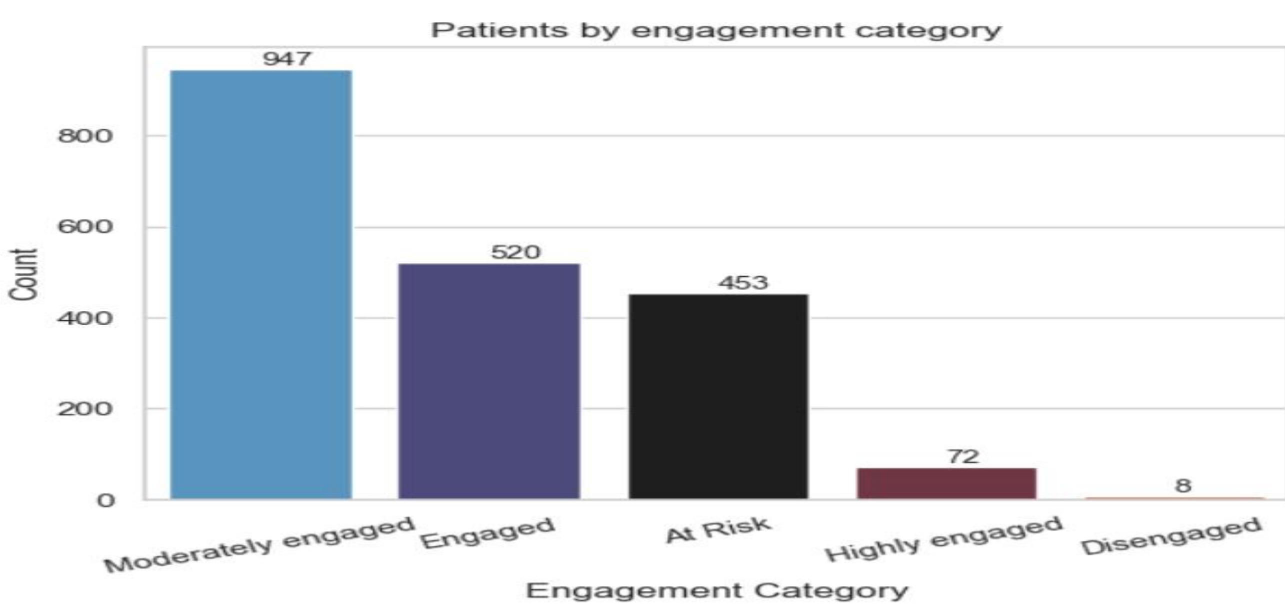
A composite engagement framework was created to combine patient activity, recency, attendance, and tenure into a single score ranging from 0 to 100.

Component	Meaning	Weight
visit_score	Normalized annual visits	35%
recency_score	Normalized inverse of days since last visit	35%
attendence_score	Higher score for fewer missed appointments	20%
tenure_score	Normalized patient tenure	10%

The engagement score was calculated as: 35% Visit Score + 35% Recency Score + 20% Attendance Score + 10% Tenure Score.

5.1 Engagement Categories

Category	Score Range	Patients
Highly engaged	80–100	72
Engaged	60–79.99	520
Moderately engaged	40–59.99	947
At Risk	20–39.99	453
Disengaged	<20	8



Insight: Most patients fall in the middle engagement tiers. The At Risk segment is particularly important because it contains 453 patients who may be suitable for proactive retention interventions.

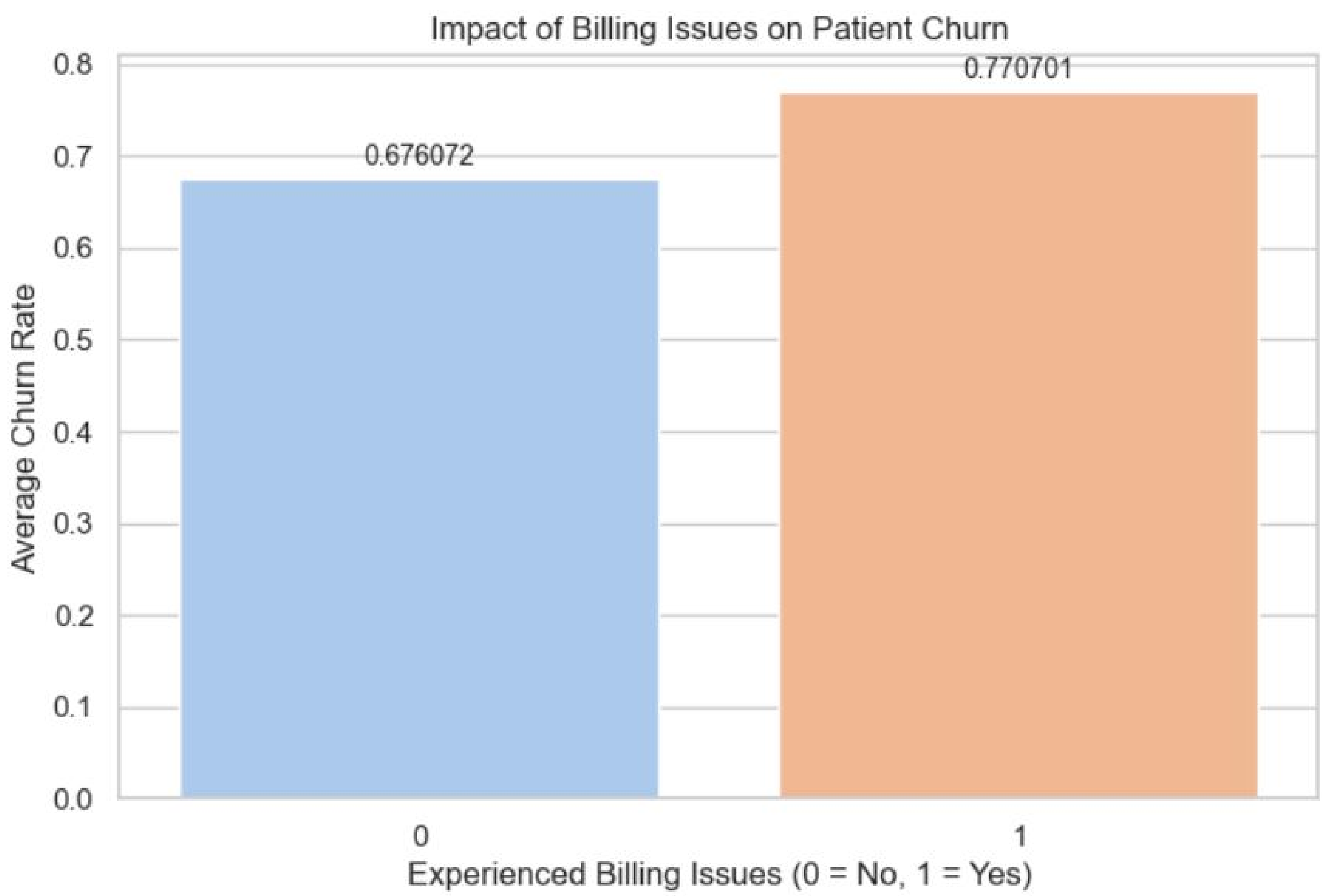
6. Exploratory Data Analysis and Insights

6.1 Patient Distribution by Medical Specialty

Specialty	Patients
General Practice	302
Family Medicine	289
Orthopedics	289
Neurology	286
Pediatrics	284
Internal Medicine	277
Cardiology	273

General Practice has the largest patient count (302), followed by Family Medicine and Orthopedics (289 each). The specialty distribution is reasonably spread across seven specialties, so the churn analysis should not rely only on the largest specialty.

6.2 Billing Issues and Churn



Patients with a recorded billing issue have an average churn rate of approximately 77%, compared with approximately 68% for patients without a billing issue. This is an observed association in the dataset, not proof that billing issues cause churn.

- Business implication: improve invoice clarity, claims communication, billing support, and issue-resolution turnaround.
- The EDA also found billing issues concentrated most heavily in the 51–65 age group (56 patients), followed by 36–50 (48).

6.3 Out-of-Pocket Cost vs Overall Satisfaction

Used a regression plot comparing average out-of-pocket cost with overall satisfaction. The fitted trendline was described as practically flat, with satisfaction remaining near 3.2 across a wide cost range. Therefore, the dataset does not show a strong linear relationship between these two variables.

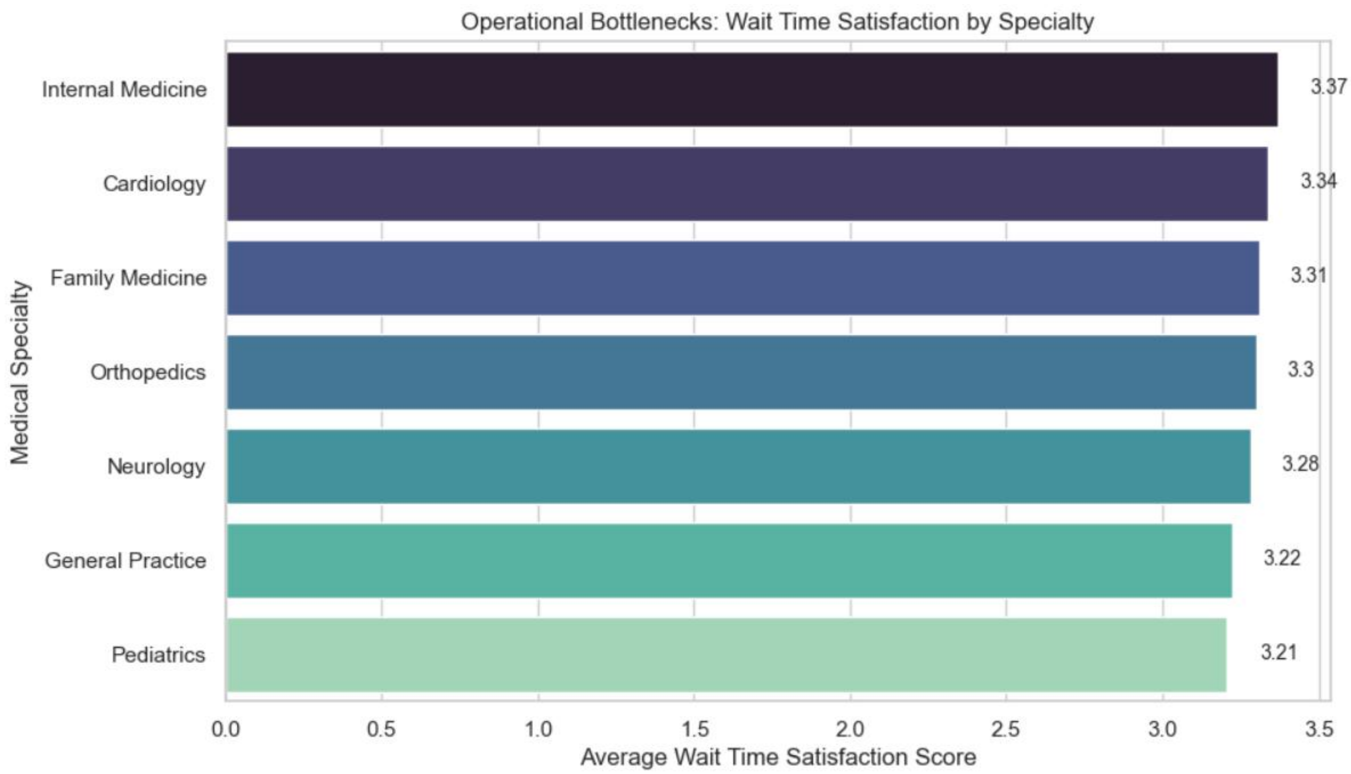
- Interpretation: patient satisfaction appears to depend on factors beyond out-of-pocket cost alone.
- Operational implication: investigate service quality, provider experience, staff interactions, convenience, and access rather than focusing only on price.

6.4 Engagement and Missed Appointments

The boxplot analysis showed that Highly engaged patients have the lowest median missed appointments (around one), while At Risk and Disengaged groups show higher absenteeism.

- Interpretation: lower engagement is associated with weaker appointment adherence.
- Business implication: engagement campaigns can be linked with appointment reminders and follow-up workflows.

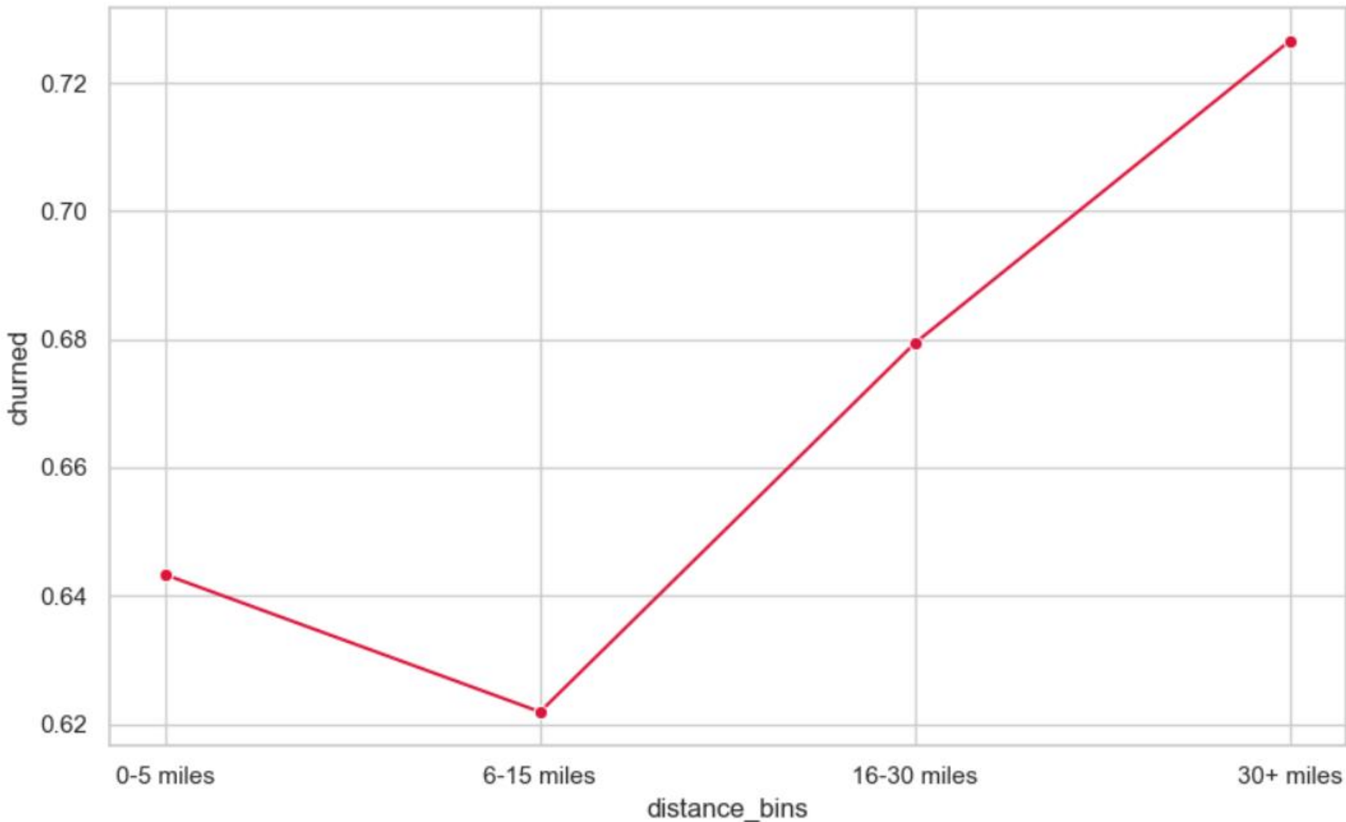
6.5 Wait-Time Satisfaction by Specialty



Specialty	Avg. Wait-Time Satisfaction
Internal Medicine	3.37
Cardiology	3.34
Family Medicine	3.31
Orthopedics	3.30
Neurology	3.28
General Practice	3.22
Pediatrics	3.21

Internal Medicine has the highest average wait-time satisfaction (3.37), while Pediatrics is lowest (3.21). The notebook interpretation notes that the differences are not large enough to identify one single problem department; improvement should therefore be considered as a clinic-wide operational initiative.

6.6 Churn by Distance



Distance	Patients	Churn Rate
0-5 miles	171	64.33%
16-30 miles	630	67.94%
30+ miles	797	72.65%
6-15 miles	402	62.19%

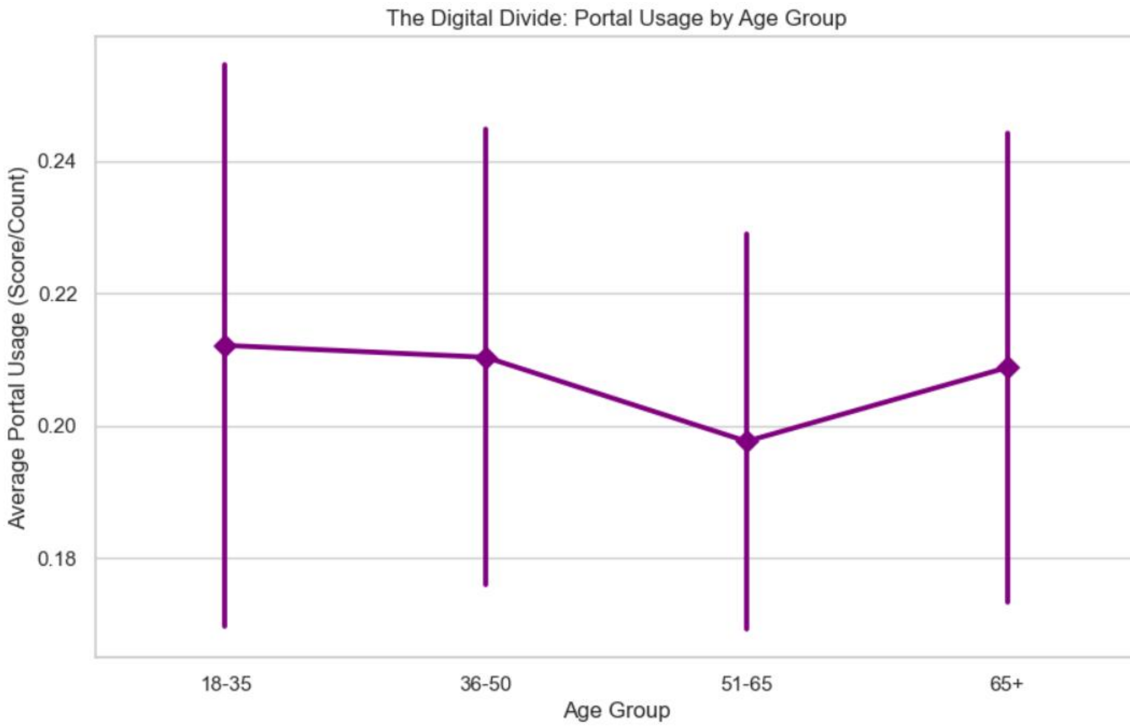
Churn is lowest for patients 6–15 miles away (62.19%) and increases substantially at longer distances, reaching 72.65% for patients more than 30 miles away. The EDA therefore identifies geographic access as one of the strongest observed churn signals.

- Business implication: consider telehealth, outreach, transportation support, regional clinics, or geographically targeted engagement for distant patients.

6.7 Portal Usage by Age Group

The notebook compared portal usage across age groups and concluded that there was no meaningful digital divide: older patients were adopting portal tools at rates comparable with younger groups.

- Implication: digital-first patient communication can be expanded without assuming that older patients will be excluded.
- The finding should be interpreted as a dataset-level observation rather than a universal statement about all patient populations.



6.8 Satisfaction Correlation Analysis

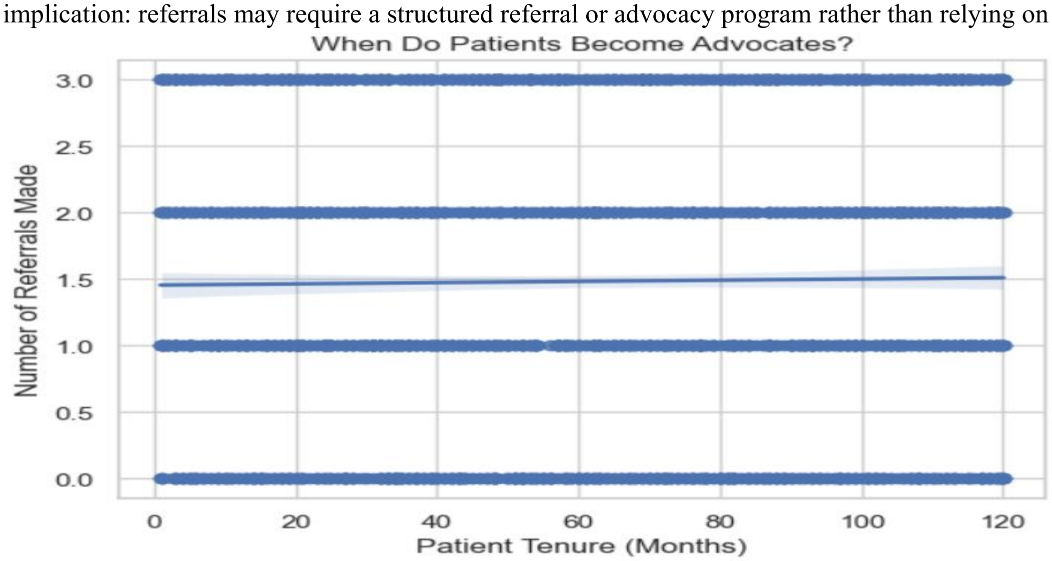
Variable	overall_satisfaction	staff_satisfaction	provider_rating	wait_time_satisfaction
overall_satisfaction	1.000	-0.025	0.013	0.036
staff_satisfaction	-0.025	1.000	0.000	-0.034
provider_rating	0.013	0.000	1.000	0.038
wait_time_satisfaction	0.036	-0.034	0.038	1.000

The notebook created a correlation heatmap for overall satisfaction, staff satisfaction, provider rating, and wait-time satisfaction. In the supplied dataset, these variables show weak pairwise linear correlations with overall satisfaction. This suggests that simple one-variable explanations of satisfaction are limited.

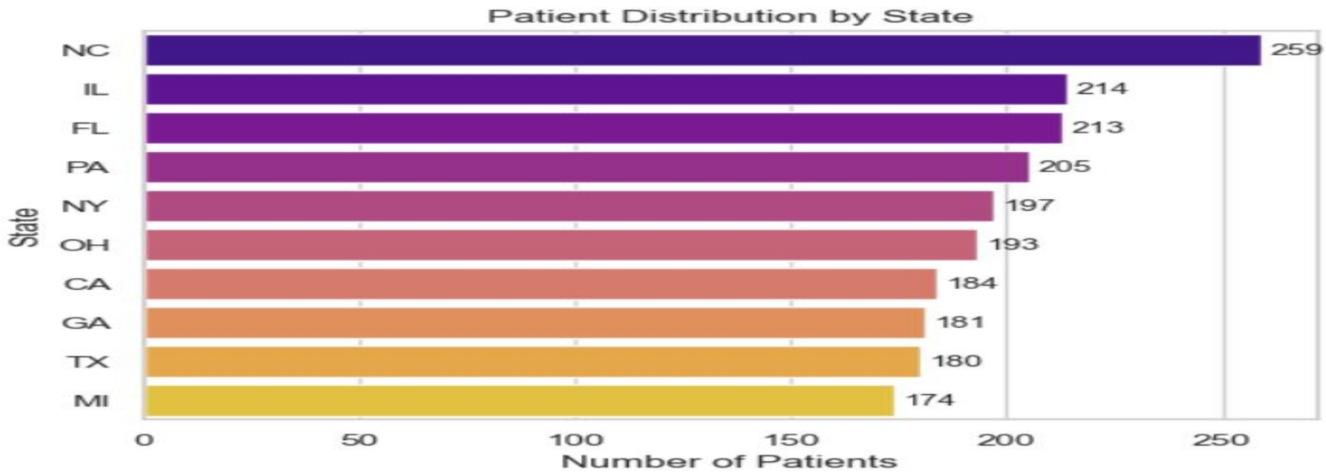
6.9 Tenure and Referrals

The notebook used a regression plot of tenure versus referrals made and concluded that long tenure did not appear to translate into proportionally higher referrals. A patient with long tenure was not necessarily more likely to refer than a patient with shorter tenure.

- Business implication: referrals may require a structured referral or advocacy program rather than relying on tenure alone.



6.10 Patient Distribution by State



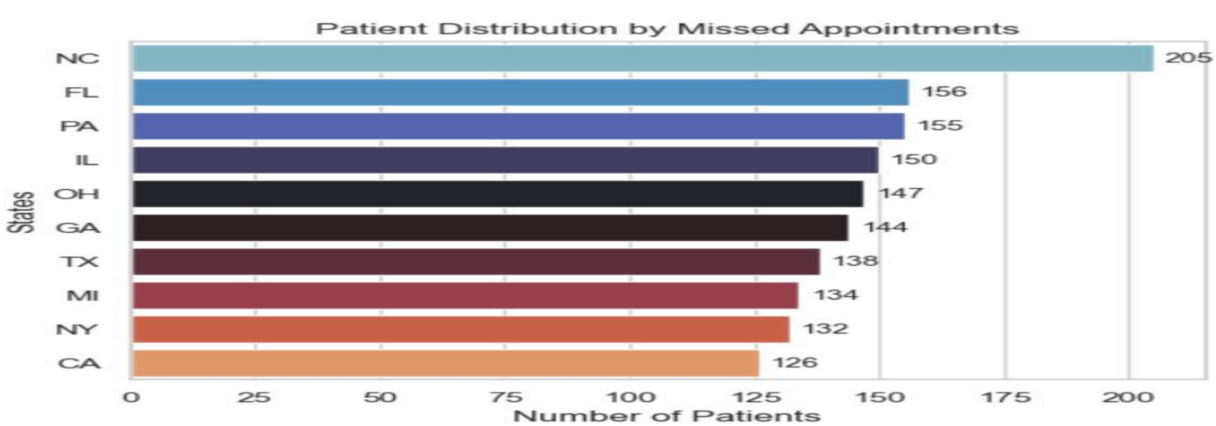
State	Patients
NC	259
IL	214
FL	213
PA	205
NY	197
OH	193
CA	184
GA	181
TX	180
MI	174

North Carolina has the largest patient count at 259, followed by Illinois (214) and Florida (213). Therefore identified NC, IL, and FL as the largest immediate volume markets.

6.11 Missed-Appointment Patients by State

State	Patients with ≥1 Missed Appointment
NC	205
FL	156
PA	155
IL	150
OH	147
GA	144
TX	138
MI	134
NY	132
CA	126

Among patients with at least one missed appointment, North Carolina has the highest count at 205, while California has the lowest at 126. Because these are counts rather than rates adjusted for each state's total patient population, they should be interpreted as workload/volume signals rather than direct evidence that one state has the highest missed-appointment rate.



6.12 Churn by Gender

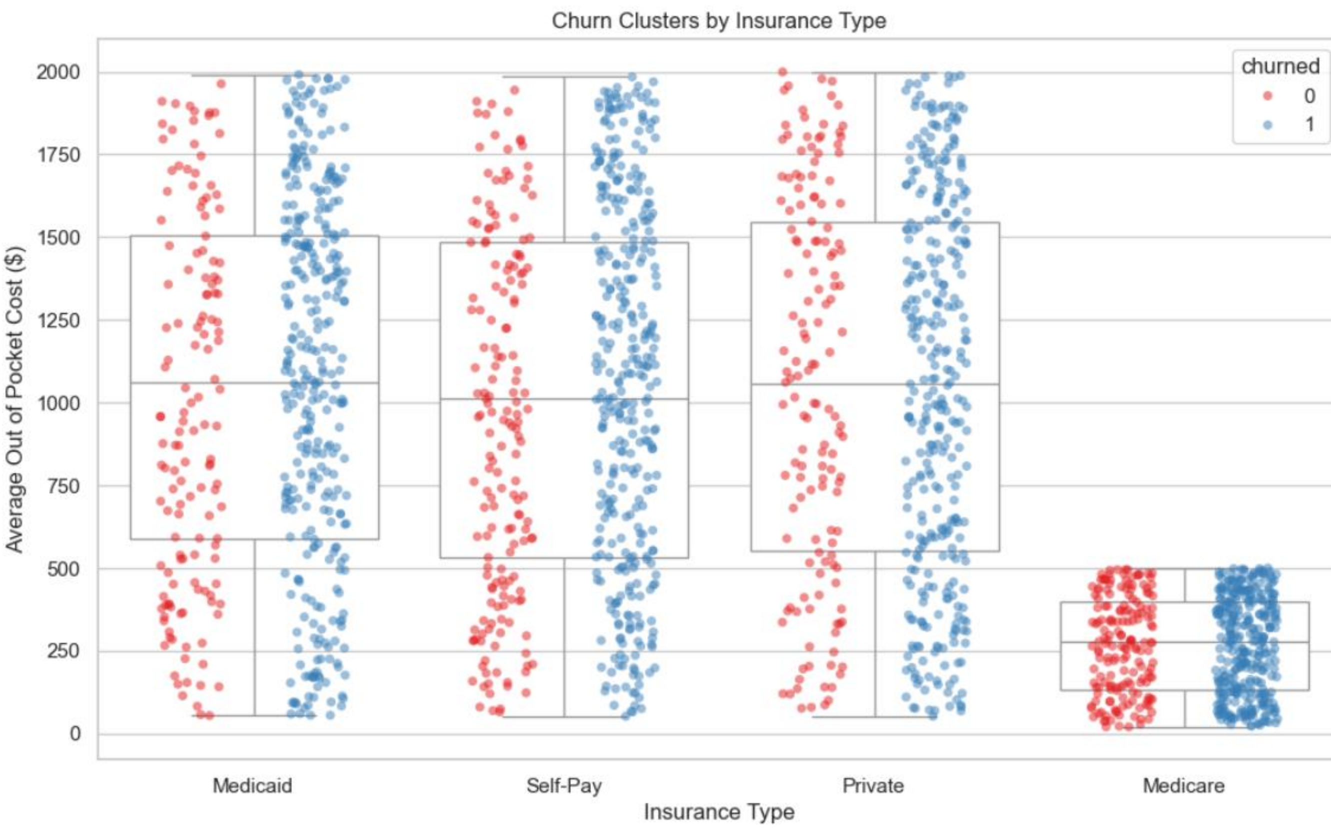
Gender	Churned Patients	Churn Rate
Female	666	68%
Male	701	69%

Male and female churn rates are very similar in the supplied analysis (69% versus 68%). Gender therefore does not appear to be a major differentiating churn factor in this dataset.

6.13 Insurance Type and Churn

Insurance Type	Patients	Churned	Churn Rate
Medicaid	478	337	70.50%
Medicare	516	344	66.67%
Private	483	335	69.36%
Self-Pay	523	351	67.11%

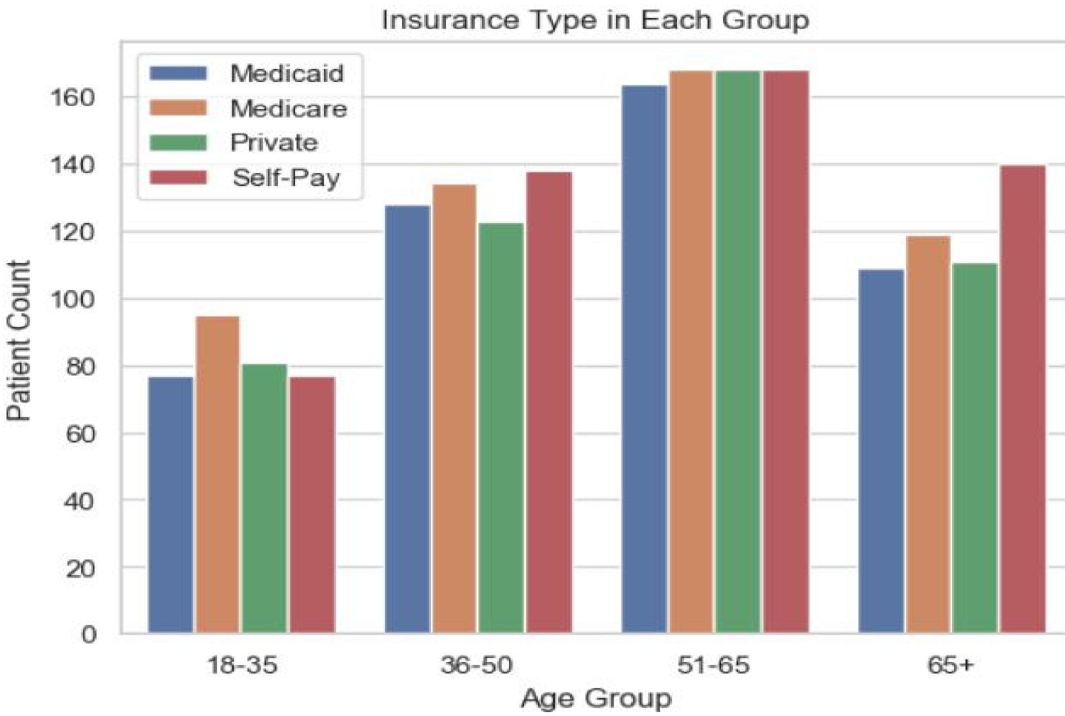
Medicaid has the highest observed churn rate at 70.50%, followed by Private insurance at 69.36%. Self-Pay and Medicare are lower at 67.11% and 66.67%, respectively. The differences are modest and should be investigated alongside other patient characteristics rather than treated as standalone causes.



6.14 Insurance Distribution by Age

Age Group	Medicaid	Medicare	Private	Self-Pay
18-35	77	95	81	77
36-50	128	134	123	138
51-65	164	168	168	168
65+	109	119	111	140

The 51–65 group is the largest age cohort and has relatively balanced representation across insurance types. The 65+ group has a comparatively higher Self-Pay count (140).



6.15 Billing Issues by Age Group

Age Group	Patients with Billing Issues
18-35	28
36-50	48
51-65	56
65+	25

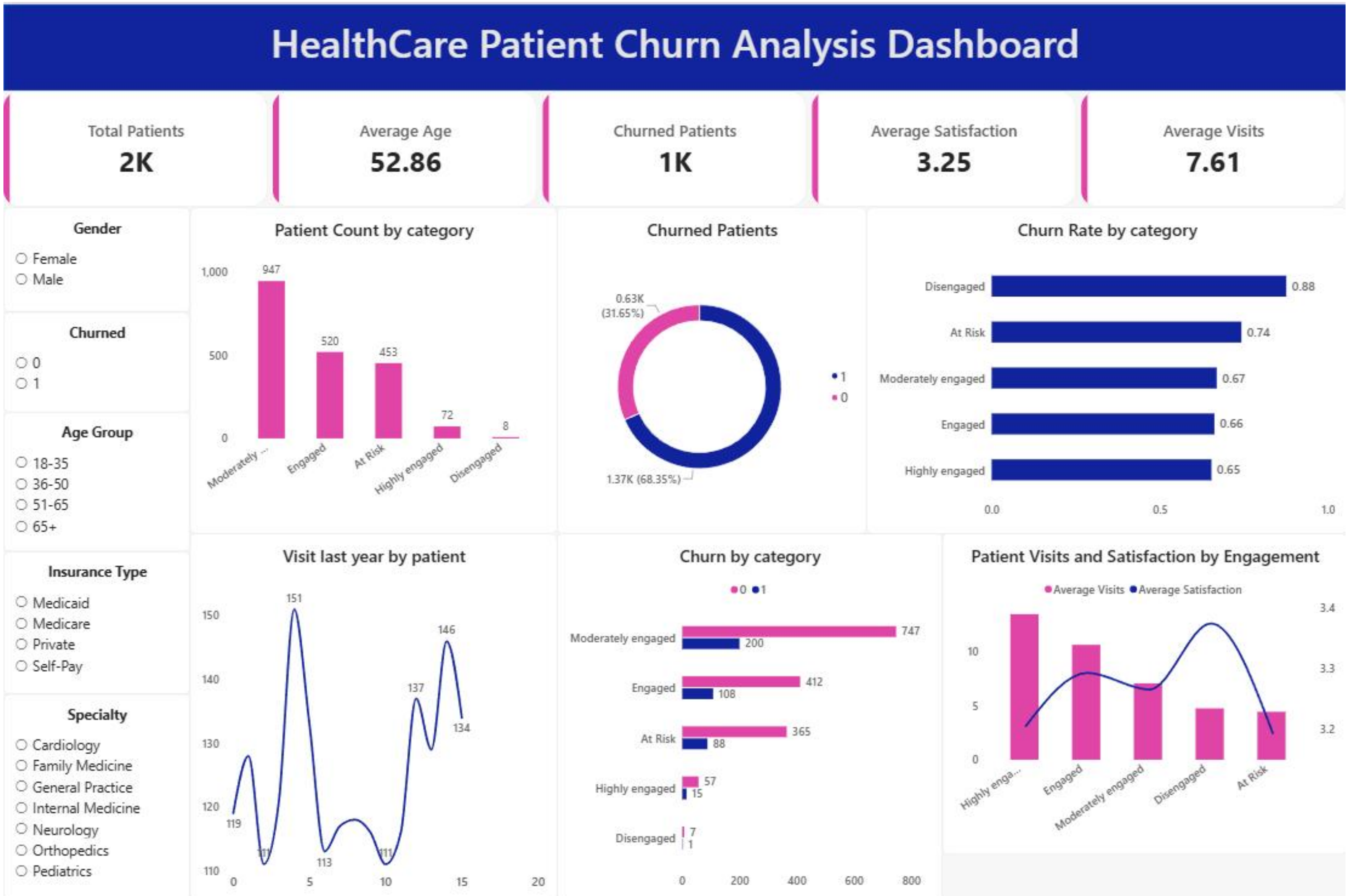
Billing issues are most frequent in the 51–65 cohort (56 patients), followed by 36–50 (48). This supports prioritizing billing-process reviews for working-age and pre-retirement cohorts.

6.16 Missed Appointments by Age Group

Age Group	Missed=0	Missed=1	Missed=2	Missed=3	Missed=4	Missed=5
18-35	85	48	53	55	47	42
36-50	129	122	78	65	77	52
51-65	179	124	125	86	87	67
65+	120	101	69	66	69	54

The age-group heatmap was used to compare the distribution of missed appointments. No single age group dominates every missed-appointment level, so appointment adherence should be managed using engagement and behavioral signals rather than age alone.

7. Power BI Dashboard



8. Business Recommendations

Prioritize the At Risk segment: Create targeted retention workflows for the 453 At Risk patients, using appointment reminders, follow-ups, and engagement campaigns.

Reduce billing friction: Audit billing errors, simplify invoices, improve claims communication, and provide faster issue resolution—especially for cohorts with higher billing-issue counts.

Address geographic access: Patients more than 30 miles away show approximately 72.65% churn; consider telehealth, outreach, transportation assistance, or regional access options.

Improve appointment adherence: Link engagement scores with reminder programs and proactive outreach to reduce missed appointments.

Use system-wide wait-time improvements: Wait-time satisfaction is relatively consistent across specialties, so operational changes should focus on clinic-wide scheduling and capacity processes.

Build a structured referral program: Tenure alone does not appear to drive referrals; introduce explicit referral/advocacy mechanisms if growth through referrals is a goal.

Expand digital engagement: The EDA did not identify a strong age-based portal adoption gap, supporting continued digital communication initiatives.

Use geography for resource allocation: NC, IL, and FL have the largest patient volumes; these markets are likely to produce the largest immediate operational impact from targeted initiatives.

9. Conclusion

The completed analysis provides a structured view of patient churn across 2,000 records. With an overall churn rate of 68.35%, the most actionable signals identified are engagement status, billing friction, geographic distance, and appointment adherence. The Power BI dashboard turns these findings into an interactive management view, while the engineered engagement score provides a practical way to segment patients for targeted retention efforts.